BIOMETRICS *29*, 121–130
March 1973

REGRESSION TOWARD THE MEAN IN UNCONTROLLED CLINICAL STUDIES

KENNETH E. JAMES

Departments of Anesthesia and Community and Preventive Medicine, Biostatistics Division, Stanford University, Stanford, California 94305

SUMMARY

When analyzing data from situations in which before and after treatment measurements have been obtained, without a parallel control group, it is important that the investigator be aware of possible changes in the observed variable due to regression toward the mean. This phenomenon can be misleading and is sometimes overlooked in the interpretation of data. This paper demonstrates the importance of a control group by distinguishing between the true treatment effect and the regression effect for the case of bivariate normal distribution, truncated with regard to the predictor or "before treatment" measurement. The regression effect is noted to be especially significant when the correlation between the before and after treatment measurement is small. A method for estimating the treatment and regression effects is presented and applied to data from an actual study.

1. INTRODUCTION

Walker ([1929] Chapter V) briefly discusses the history of regression toward the mean. She states that Bravais, Bowditch, and others considered the problems of correlation and regression from a mathematical point of view. However, the practical significance of the formulas which they derived did not seem apparent to them.

Galton [1886] considered the regression problem when experimenting with the growth of peas. He observed that the mating of two tall plants produced offspring which were, on the average, shorter than either parent plant. Likewise, the mating of two short plants produced offspring which were taller than either parent. Galton also observed this phenomenon in humans and was able to construct a "forecaster" with which he could predict the stature of offspring of certain parents whose heights were known to him. Mathematically, the "forecaster" which Galton used was the regression line of y on x for the bivariate normal distribution, where x denotes the mid-parental height of the father and mother and y the height of their offspring. The regression equation is stated in the form

$$E(y \mid x) = \mu_y + \rho \frac{\sigma_y}{\sigma_x} (x - \mu_x),$$

where $\rho(\sigma_y/\sigma_x)$ is the slope of the line. In most genetic populations, $\sigma_y = \sigma_x$ and $\mu_y = \mu_x$ are expected for the bivariate distribution. Making these assumptions, the slope of the regression line is equal to ρ, and, since $|\rho| < 1$, it is evident that the height of the offspring will tend toward the mean value μ_y . Pearson ([1930] Chapter XIV) discusses many of the aspects of this approach in his biography of Galton.

In applying these concepts to clinical investigations involving measurements taken before and after the application of a treatment, measurement error must also be considered; i.e. it is assumed that the measurement of a clinical variable is subject to a within person component of variability. Furthermore, it should be realized that this variability is subject to a regression effect. Thus two regression effects are present in the before-after measurement situation: that due to biological variation over time, and the effect due to the variation of the measurement. In the clinical situation, therefore, it is evident that if there were either day-to-day variations within patients, or if there were measurement errors involved in the individual determinations, Galton's regression toward the mean would be seen, whether or not the treatment itself were effective. Note that here and in the remainder of the paper the term "treatment" refers to the actual application of a drug or medical procedure to a patient. Considering as an example before-after measurements of blood pressure, patients with high initial readings will show a decrease on the average, while those with low readings will show an increase. Of course, in a clinical investigation of a treatment for high blood pressure, patients are screened before treatment and only those with high initial values are treated. Thus the final data will resemble a truncated sample from a bivariate population. If the screened patients exhibit a decrease in blood pressure following treatment, it will be difficult to decide whether the decrease is completely explainable in terms of the regression effect, or whether it is partly due to the effect of the treatment. This problem is examined in some detail in the following sections.

It should be emphasized here that this paper is not written to encourage uncontrolled clinical trials, but rather to underline one of the many difficulties in interpreting the results of the uncontrolled clinical study.

2. THE NULL CASE

Let the random variables x and y represent respectively the patient's measurements before and after treatment. We assume that the measurements estimate a true value u with errors w_1 and w_2 respectively. That is, $x = u + w_1$ and $y = u + w_2$, where u is distributed $N(\mu, \sigma_u^2)$, w_1 , w_2 are independently identically distributed $N(0, \sigma_w^2)$ and are independent of u. Thus x and y are both distributed $N(\mu, \sigma^2)$ where $\sigma^2 = \sigma_u^2 + \sigma_w^2$, and $\rho = \text{corr}(x, y) = \sigma_u^2/\sigma^2$. Only patients with values above a truncation point x_0 are treated and measured for treatment effect. Finney [1956] uses characteristic functions to obtain the expected values and variances for x and y in the truncated sample. An alternate method involves the determination of certain combinations of $E(x^r y^s \mid x \geq x_0)$ by evaluating the integral

$$\frac{1}{1-\Phi(x_0)} \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{x_0}^{\infty} \int_{-\infty}^{\infty} x^r y^s \exp\left[-(x^2 - 2\rho xy + y^2)/2(1-\rho^2)\right] dy\, dx$$

for r, $s = 0, 1, 2$, where Φ denotes the c.d.f. for the normal distribution. Var $(d \mid x \geq x_0)$, where $d = y - x$, is easily obtained by combining Var $(y \mid x \geq x_0)$ and Var $(x \mid x \geq x_0)$ and noting that cov $(x, y \mid x \geq x_0) = \rho$ Var $(x \mid x \geq x_0)$. Table 1 shows the results for x, y and d.

TABLE 1

MEAN AND VARIANCE OF THE DISTRIBUTIONS OF THE BEFORE MEASUREMENT (x), THE AFTER MEASUREMENT (y), AND THE DIFFERENCE ($d = y - x$), ASSUMING TRUNCATION ON THE BEFORE MEASUREMENT ($x \geq x_0$)

	Pre-treatment $(x \mid x \geq x_0)$	Post-treatment $(y \mid x \geq x_0)$	Difference $(d = y-x \mid x \geq x_0)$
Mean	$\mu + k_0\sigma$	$\mu + \rho k_0\sigma$	$k_0\,\sigma(\rho-1)$
Variance	$\sigma^2[k_0(z_0-k_0)+1]$	$\sigma^2[\rho^2 k_0(z_0-k_0)+1]$	$(1-\rho)[k_0(z_0-k_0)(1-\rho)+2]\sigma^2$

where

μ = unconditional mean of both x and y

σ = unconditional standard deviation of both x and y

$z_0 = (x_0 - \mu)/\sigma$

$k_0 = \phi(z_0)/[1 - \Phi(z_0)]$

ϕ, Φ are the standardized normal density and distribution functions respectively

Table 2 shows numerical results for the standardized bivariate normal distribution with varying values of ρ and x_0. Note the large decrease in the mean due to the regression effect alone when the correlation is small.

3. THE NONNULL CASE

With *no* treatment effect, it is well known that the post-treatment measurements, y, can be written as

$$y = \rho x + e_1$$

TABLE 2

MEAN AND VARIANCE OF THE STANDARDIZED TRUNCATED BIVARIATE DISTRIBUTION

$\Phi(x_0)$	x_0	ρ	Pre-treatment measurement (x; $x \geq x_0$)		Post-treatment measurement (y; $x \geq x_0$)		Differences ($d = y-x$; $x \geq x_0$)	
			Mean	Variance	Mean	Variance	Mean	Variance
0.75	0.67	0.2	1.27	0.2380	0.254	0.9695	-1.016	1.1123
0.75	0.67	0.5	1.27	0.2380	0.635	0.8095	-0.635	0.8095
0.75	0.67	0.8	1.27	0.2380	1.016	0.5123	-0.254	0.3695
0.80	0.84	0.2	1.40	0.2160	0.280	0.9686	-1.120	1.0982
0.80	0.84	0.5	1.40	0.2160	0.700	0.8040	-0.700	0.8040
0.80	0.84	0.8	1.40	0.2160	1.120	0.4982	-0.280	0.3686
0.85	1.04	0.2	1.56	0.1888	0.312	0.9676	-1.248	1.8080
0.85	1.04	0.5	1.56	0.1888	0.780	0.7972	-0.780	0.7972
0.85	1.04	0.8	1.56	0.1888	1.248	0.4808	-0.312	0.3676

where

$$E(e_1 \mid x) = 0$$

$$\text{Var}\ (e_1 \mid x) = 1 - \rho^2,$$

for the standard bivariate normal distribution.

Now consider a very simple kind of treatment effect, but one that is reasonable for types of clinical studies of interest here. If the treatment is effective it is assumed to alter the post-treatment measurements, so that they have the form:

$$y' - \mu = \gamma\rho(x - \mu) + e_2$$

where $|\gamma| < 1$. In other words we consider a treatment designed to move the measurements toward the pre-treatment mean (μ) of the untruncated population. The treatment effect is stated in terms of the pre-treatment measurement, x, and a measurement error e_2 which has the same properties as e_1 given above.

For γ nonzero, the properties of the distributions of y' and d', for the standard bivariate distribution truncated at x_0, are easily obtained:

$$\begin{aligned} E(y' \mid x \geq x_0) &= \gamma\rho E(x \mid x \geq x_0) \\ &= \gamma\rho k_0 \\ \text{Var}\ (y' \mid x \geq x_0) &= \gamma^2\rho^2\ \text{Var}\ (x \mid x \geq x_0) + (1 - \rho^2) \\ &= \gamma^2\rho^2[k_0(x_0 - k_0) + 1] + (1 - \rho^2) \end{aligned}$$

$$E(d' \mid x \geq x_0) = E[(y' - x) \mid x \geq x_0]$$
$$= k_0(\gamma\rho - 1)$$
$$\mathrm{Var}\,(d' \mid x \geq x_0) = \mathrm{Var}\,[(y' - x) \mid x \geq x_0]]$$
$$= (\gamma\rho - 1)^2[k_0(x_0 - k_0) + 1] + (1 - \rho^2).$$

Numerical results for the standard bivariate normal distribution with $\rho = 0.5$, $x_0 = 0.67$, and $\gamma = 0.2$ are presented in Table 3.

TABLE 3

DISTRIBUTION OF y' AND d' (NONNULL CASE) FOR THE STANDARD BIVARIATE NORMAL WITH TREATMENT EFFECT $\gamma = 0.2$

	Distribution of x Pre-treatment	Distribution of y' Post-treatment		Distribution of d' Post-treatment	
		$\gamma = 1$	$\gamma = 0.2$	$\gamma = 1$	$\gamma = 0.2$
Mean	1.27	0.635	0.127	-0.635	-1.14
Variance	0.2380	0.8095	0.7824	0.8095	0.9428

Note that the proportional reduction in the mean at level x, due to both regression effect and treatment effect is:

$$\text{total proportion reduction} = (x - \gamma\rho x)/x = 1 - \gamma\rho.$$

The proportion reduction that would be observed due to regression alone would be:

$$\text{proportion reduction due to regression} = 1 - \rho.$$

Thus, the proportion of the reduction due to regression relative to the total reduction, is:

$$\frac{\text{proportion reduction due to regression}}{\text{total proportion reduction}} = (1 - \rho)/(1 - \gamma\rho).$$

Hence, if ρ is close to zero almost all of the reduction observed would have been observed without the benefit of any effective treatment.

As an example of the relative importance of the regression and treatment effects in the reduction of the mean, Table 4 is presented. Note that the ratio of [reduction due to regression] to [reduction due to treatment] ranges from 2:1 to 1:7, depending on the value of ρ. Clearly the reduction due to regression comprises a substantial amount of the total reduction, even when ρ is high.

TABLE 4

PERCENT REDUCTION IN THE MEAN DUE TO REGRESSION AND TREATMENT EFFECTS

		Percentage reduction in the mean		
ρ	γ	Due to regression alone	Remainder due to treatment	Total
0.4	0.1	60	36	96
0.4	0.2	60	32	92
0.4	0.3	60	28	88
0.5	0.2	50	40	90
0.5	0.3	50	35	85
0.6	0.2	40	48	88
0.6	0.4	40	36	76
0.8	0.2	20	64	84
0.8	0.4	20	48	68
0.8	0.5	20	40	60
0.9	0.2	10	72	82
0.9	0.4	10	54	64
0.9	0.6	10	36	46
0.9	0.7	10	27	37

4. ESTIMATION OF THE PARAMETERS FROM SAMPLE DATA

For the bivariate regression situation, the first two moments of x and y plus the slope of the observed best fitting line may be used to estimate μ, σ^2, ρ and γ. We consider the percent of the population in the truncated portion to be known; equivalently, k_0 is taken to be known. Although in practice k_0 may not be known exactly, it is reasonable to assume that the investigator is familiar enough with his population of subjects to be able to provide a suitable estimate. By the method of moments, and referring to Table 1, we have:

$$b_{y'\cdot x} = \hat{\gamma}\hat{\rho}, \quad \text{hereafter referred to as simply } b \tag{3.1}$$

$$\bar{x} = \hat{\mu} + k_0\hat{\sigma}, \quad s_x^2 = \hat{\sigma}^2(k_0(x_0 - k_0) + 1), \tag{3.2}$$

$$s_{y'}^2 = \hat{\sigma}^2[\hat{\gamma}^2\hat{\rho}^2 k_0(x_0 - k_0) + \hat{\gamma}^2\hat{\rho}^2 + (1 - \hat{\rho}^2)]. \tag{3.3}$$

From (3.2)

$$\hat{\mu} = \bar{x} - k_0\hat{\sigma} \quad \text{and} \quad \hat{\sigma}^2 = \frac{s_x^2}{(k_0(x_0 - k_0) + 1)}.$$

From (3.3)

$$\hat{\rho} = \left[b^2(k_0(x_0 - k_0) + 1) - \frac{s_{y'}^2}{\hat{\sigma}^2} + 1\right]^{1/2}, \tag{3.4}$$

and from (3.1)

$$\hat{\gamma} = \frac{b}{\hat{\rho}}.$$

Variances of the estimates are obtained as follows:

$$\text{Var}\,(\hat{\sigma}^2) = \frac{2\sigma_x^4}{(k_0(x_0 - k_0) + 1)^2(n - 1)}.$$

Using Taylor series approximations,

$$\text{Var}\,(\hat{\sigma}) \doteq \frac{\sigma_x^4}{2\sigma^2(k_0(x_0 - k_0) + 1)^2(n - 1)}$$

from which

$$\begin{aligned}\text{Var}\,(\hat{\mu}) &= \text{Var}\,(\bar{x}) + k^2\,\text{Var}\,(\hat{\sigma}) \\ &\doteq \frac{\sigma_x^2}{n} + \frac{k_0^2\sigma_x^4}{2\sigma^2(k_0(x_0 - k_0) + 1)^2(n - 1)}.\end{aligned}$$

Using Taylor's series approximations for two variables, it can be shown that

$$\begin{aligned}\text{Var}\,(\hat{\rho}) \doteq{}& \frac{(k_0(x_0 - k_0) + 1)^2\gamma^2\sigma^2}{\sum (x_i - \bar{x})^2} + \frac{\sigma_{y'}^4}{2\rho^2\sigma^4(n - 1)} \\ &+ \frac{\sigma_x^4\sigma_{y'}^4}{2\rho^2\sigma^8(k_0(x_0 - k_0) + 1)^2(n - 1)}\end{aligned}$$

and

$$\begin{aligned}\text{Var}\,(\hat{\gamma}) ={}& \frac{\sigma^2}{\rho^2\sum (x_i - \bar{x})^2} + \frac{(k_0(x_0 - k_0) + 1)\gamma^4\sigma^2}{\rho^2\sum (x_i - \bar{x})^2} \\ &+ \frac{\gamma^2\sigma_{y'}^4}{2\rho^4\sigma^4(n - 1)} + \frac{\gamma^2\sigma_x^4\sigma_{y'}^4}{2\rho^4\sigma^8(k_0(x_0 - k_0) + 1)^2(n - 1)}.\end{aligned}$$

5. A NUMERICAL EXAMPLE

In a study performed to ascertain the effect of surgery for high blood pressure thought to be due to renovascular pathology, the following data from 341 patients were obtained on diastolic blood pressure before and after the operation:

$$\bar{x} = 110.1 \qquad \bar{y}' = 89.7$$
$$s_x = 13.1 \qquad s_{y'} = 13.6$$
$$b_{y'\cdot x} = 0.265$$

Before entering the study, patients were screened on the basis of their past medical history. Among those screened, a certain proportion were subjected to surgery on the basis of their pre-treatment blood pressure levels. This proportion is assumed to be 25% and the distribution of the screened population will be taken to be bivariate normal. Of course, these assumptions are simplifications, made for the sake of illustration, and should not be taken as a serious or definitive analysis of the data at hand.

Table 5 shows the numerical estimates and the respective estimates of variances for μ, σ^2, ρ and γ.

TABLE 5
NUMERICAL ESTIMATES OBTAINED FROM A BLOOD PRESSURE STUDY

	Parameters			
	μ	σ^2	ρ	γ
Estimates	76.0	720.8	0.87	0.30
Var. of estimates	2.21	1528.0	0.0032	0.0164
S.E. of estimates	1.49	39.09	0.018	0.128

Note that in section 4 the first moment of y' was not used in the estimation of μ, σ, γ, or ρ. However, substitution of the relevant parameters into the equation

$$\bar{y}' = \hat{\mu} + \hat{\gamma}\hat{\rho}k_0\hat{\sigma}$$

yields a value of 85.0, which is in good agreement with the observed $\bar{y}'$ of 89.7, though the discrepancy is statistically significant in the light of the standard error of 0.74 for $\bar{y}'$. Thus, though the data seems to serve reasonably well as an example for our purposes, improvements in the model might be desirable.

The analysis indicates that without the surgery blood pressures of x would be reduced to an average of $\mu + \rho(x - \mu)$ or $76.0 + (0.87)(x - 76.0)$, and that the surgery reduced the pressures further to $\mu + \gamma\rho(x - \mu)$ or $76.0 + 0.27\ (x - 76.0)$. Thus blood pressures of 120 would later average 114.3 without the operative procedure and 87.9 with the operative procedure, a substantial real decrease in diastolic pressure due to the surgery.

6. DISCUSSION

In the performance of a clinical study designed to ascertain the effect of a certain treatment, the use of a control group is usually wise, ethics permitting. This permits the direct correction for the regression effect by noting the reduction in the mean of the controls. The regression effect, and thus the importance of a control group, becomes critical when the correlation between the before and after treatment measurements is small. When the statistician is confronted with study results for which control measurements are not available, some variation of the procedure outlined may be a means of separating the regression and treatment effects; but the existence of such techniques should not be used to argue for the elimination of control measurements. To the contrary, the difficulties in these analyses argue strongly for randomized clinical experiments.

The estimation procedures presented here are based on the method of moments and may be far from best for small samples. Furthermore, the truncation percentage was taken to be known, and this assumption must always be carefully checked in any real application.

Other methods of parameter estimation and robustness of the estimates to nonnormality are still very much open to further investigation.

ACKNOWLEDGMENTS

I wish to extend my appreciation to Professor Byron W. Brown, Jr. for his thoughtful suggestions and guidance in the preparation of this paper for publication. I also wish to thank Mr. Paul Varady of Cedars-Sinai Medical Center, Los Angeles, and the investigators in the Cooperative Reno-vascular Supertension Study, for supplying the data which was used in section 5. This work was first presented as a paper in partial fulfillment of the requirements for the M.S. in Biostatistics at the University of Minnesota and supported on N.I.H. Training Grant 2TI-GM-43-13.

REGRESSION VERS LA MOYENNE DANS DES ETUDES CLINIQUES EN L'ABSENCE D'UN GROUPE TEMOIN

RESUME

Quand on analyse des données dans des situations telles que des mesures ont été obtenues avant et après un traitement, et en l'absence d'un groupe témoin parallèle, il est important que le chercheur soit conscient des modifications possibles de la variable observée, du fait d'une régression vers la moyenne. Ce phénomène peut entrainer des erreurs et est souvent négligé dans l'interprétation des données. Le présent article démontre l'importance d'un groupe témoin en distinguant de l'effet du vrai traitement l'effet de la régression, dans le cas d'une distribution normale à deux variables, tronquée en ce qui regarde la prévision de la mesure "avant traitement". On remarque que l'effet dû à la régression est particulièrement significatif quand la corrélation entre les mesures avant et après traitement est faible. On présente une méthode en vue d'estimer les effets du traitement et de la régression et on l'applique aux données d'une étude réelle.

REFERENCES

Finney, D. J. [1956]. The consequences of selection for a variate subject to errors of measurement. *Rev. Inst. Int. Statist. 24*, 1–10.

Galton, F. [1886]. Regression towards mediocrity in hereditary stature. *Journal of the Anthropological Institute 15*, 246–63.

Pearson, K. [1930]. *Life of Francis Galton.* Cambridge University Press, London.

Walker, H. M. [1929]. *Studies in the History of Statistical Methods.* Williams and Wilkens, Baltimore.

Received June 1971, *Revised June* 1972

Key words: Regression toward the mean; Uncontrolled clinical trials; Use of correlation to show importance of a control group.